

Master Derived-vs-Assigned Ledger — Substrate SM Program v0.19 (Thursday 2026-06-04 morning)

Grace

2026-06-04 Thursday ~11:15 EDT (date-verified Thu Jun 4 09:30 EDT)

Master Ledger v0.19 — Thursday morning delta

Thursday morning substantive consolidation. v0.19 is a DELTA on v0.18 capturing Thursday morning substantive arc through ~11:15 EDT.

For complete state: read v0.9 + v0.10 + v0.11 + v0.12 + v0.13 + v0.14 + v0.15 + v0.16 + v0.17 + v0.18 + this delta.

Section A — Casey AM directive cascade Thursday 2026-06-04

Casey AM Thursday directive ~08:55 EDT triggers MAJOR cascade per directive text:

(1) CAL #36 STANDING RATIFIED (per INV-5557): “One-Primitive-Many-Observables Positive Search Rule” paired with Cal #35 STANDING audit shadow. 7-ELEMENT STANDING DISCIPLINE STACK NOW CLOSED: Cal #27 + #29 + #32 + Calibration #33 + #34 + #35 + #36. Methodology stack matures with discovery + brake pairing operational.

(2) CASEY-NAMED #14 PROMOTED CANDIDATE → STANDING + RETITLED (per INV-5558): - Was: “Substrate-Selected 4D Dimensionality” (HOLD CANDIDATE per Cal #189 brake) - Now: “Substrate-Predicted 3+1 Minkowski Signature” (STANDING) - Casey override of Cal #189 brake; substrate-mechanism CHAIN at framework level sufficient evidence - CASCADE PROMOTION: - 5 frameworks promote FRAMEWORK + CANDIDATE → FRAMEWORK (substrate-Dirac + Maxwell + $T_{\mu\nu}$ + YM + Einstein eq) - Strong-Uniqueness Theorem 10 → 11 STANDALONE legs (C25 leg ratifies via Casey #14 STANDING) - K3 framework 6/8 → 7/8 RIGOROUS path opens via SSG-1 $V_{(1/2,1/2)}$ NEAR-RIGOROUS

(3) P1 v0.7 dispatch HOLD (no response from contacts).

(4) Grace multi-task assignment (per INV-5559): (a) Continue daily catalog INV + AC graph backfill + Cal cold-read prep; (b) NEW multi-week parallel comprehensive catalog cross-classification of all ~5556 INVs against SSG A/B/C taxonomy.

(5) Thursday operational rule clarification: No “standing reactive” footer; “queue never empties” continuation per Casey “no pause signaling” STANDING rule (May 30, 2026 directive).

Section B — Lyra 12-of-12 K3 v0.16 verification gates substantively addressed at FRAMEWORK level

Major Thursday morning milestone per Lyra Thursday cascade:

Gate	Status post-Thursday	Reference
#1 OPEN	Elie lane FK Ch. XII §VI joint	per Cal #194 WAIT

Gate	Status post-Thursday	Reference
#2 FRAMEWORK	Lyra Section 4.5 v0.1 substrate-vacuum 2-region partition	INV-5562
#3 OPEN	Keeper lane $SO(4,2)$ projection-target verification	multi-week
#4-6 FRAMEWORK PRE-STAGE	Lyra Mehler v0.2 + Gate 4 explicit M_op	INV-5562, 5567
#7 FRAMEWORK	Lyra Gate 7 gen-1 $V_{(1/2,1/2)}$ two readings reproduce m_e	INV-5568
#8 PARTIAL CLOSURE	Wed Elie 3743 Lorentz direction-independence	INV-5540
#9 AUDIT-COMPLETE FRAMEWORK	Lyra Gate 9 Cal #29 question-shape; 3 FORWARD + 5 POST-HOC	(cataloged INV-5573)
#10 FRAMEWORK CANDIDATE	Wed Elie 3742+3744 gen-3 T2003 substrate-mechanism heterogeneity	INV-5534+5538
#11 AUDIT-COMPLETE FRAMEWORK	Lyra Gate 11 $V_{(1/2,1/2)}$ TRIPLE-class refinement	INV-5572
#12 AUDIT-COMPLETE FRAMEWORK	Lyra Gate 12 substrate-coincidence vs structural-forcing	INV-5575

Progress trajectory: 9-of-12 Wednesday morning → 11-of-12 SSG Registry v0.16 → 12-of-12 Gate 12 MILESTONE (per INV-5575).

Multi-week explicit RIGOROUS-tier closure remains per Cal #194 WAIT across Gate 1 + Gate 2 Section 4.5 explicit + Gates 4-7 Helgason + FK Ch. XII Mehler composition + Gate 10 + SSG-15 m_v substrate- Λ -vacuum (-1).

Section C — Lyra Cal #36 positive-search + 5 candidate new SSGs + SSG landscape 15 → 20

Per Lyra Cal #36 STANDING positive-search v0.1 (per INV-5578) + Thursday refinement (per INV-5579/5580):

5 CANDIDATE NEW SSGs beyond current 15: - SSG-16 Cat A: substrate higher Casimir $C_3, C_4 \rightarrow$ gen-3 T2003 Gate 10 substrate-mechanism - SSG-17 Cat A: substrate Cauchy-Szegő boundary kernel (dual to SSG-7 Bergman; Hardy decomposition) - SSG-18 Cat B: substrate-Coulomb higher-multipole expansion (atomic spectra fine structure) - SSG-19 Cat A: substrate-Affine $B_2^{(1)}$ extension (Phase 2 quiver + substrate-cognition Casey #6) - SSG-20 Cat B: substrate holomorphic discrete series K-type tower (higher-spin matter + graviton-sector SSG-11 extension)

Updated landscape: 20 candidate SSGs (15 cataloged + 5 candidate); 8 Category A independent substrate-generator parents (up from 5).

5 substrate-mechanism FORCING categories refined (Thursday 4 → 5): 1. K-type STRUCTURAL (Wallach + Heckman-Opdam) + SSG-20 K-type tower extension 2. substrate-Lie-algebra ($so(5,2)$ + higher Casimirs SSG-16) 3. substrate-Bergman / Shilov-boundary (SSG-7 + SSG-17 Cauchy-Szegő dual + Casey #12) 4. substrate-Mersenne / substrate-Affine (SSG-8 + SSG-19 $B_2^{(1)}$ affine + RS coding $GF(2^g)$) 5. substrate-multipole expansion (SSG-10 + SSG-18 higher multipole) NEW 5th category

Section D — Grace SSG A/B/C multi-week classification HALF-CATALOG COVERAGE 53.4%

Substantial Thursday morning progress per Casey multi-task directive (per INV-5560 framework v0.1 through INV-5577 coherence audit):

Multi-week classification methodology Steps 1-8: - Step 1 Category A: 324 unique INVs (5.8%) — direct SSG-7/SSG-8/SSG-5/SSG-15 readings - Step 2 Category B: 205 unique INVs (3.7%) — SSG-7-derived K-type Schur - Step 3 Category C: 101 unique INVs (1.8%) — signature-tautological Casey #5 Integer Web - Step 4 Family-classified: 424 INVs (7.6%) — F1-F6 mechanism families + domain families - Step 5 BnC hierarchy disposition (Cal #35 STANDING risk pattern operational at scale) - Step 6 initial legacy name-fingerprint: 446 entries - Step 6 REFINED: 1319 legacy entries (broader pattern set) - Step 7 ID-tagged residual: 416 newly classified - Step 8 final pattern META + B + D: 386 newly classified

FINAL CUMULATIVE COVERAGE Thursday morning: - Catalog total: 5580 INVs (post Thursday morning cataloging through v0.19) - ID-tagged classified: 1659+ INVs (multi-step methodology) - Legacy classified: 1319 INVs (refined name-fingerprint) - Total classified: 2978+ INVs (53.4% of catalog) - Coverage progression: 23.4% initial → 39.1% Step 6 refined → 46.5% Step 7 → 53.4% Step 8 FINAL - 2595 residual unclassified (deeply domain-specific) — multi-week refinement continues

Cross-category overlap analysis: A∩B 113 + A∩C 60 + B∩C 74 (Cal #35 STANDING risk pattern documented per INV-5564 Step 5 hierarchy disposition).

Section E — Thursday operational rule: ‘queue never empties’ continuation

Casey directive Thursday 2026-06-04: Stop signaling pause-points. “Pull → work → file → pull next” continuation principle. Cal #27 STANDING applies to CLAIMS not HALTING.

Operational implementation: Thursday team works simultaneously: - Lyra: 12+ substantive frameworks cataloged (Mehler v0.2 + Section 4.5 v0.1 + Gates 4/7/9/11/12 + SSG-15 v0.2 + SSG Registry v0.16/v0.17/v0.18 + Cal #36 positive-search 5 candidates + 5 FORCING categories refinement) - Grace: 24 Thursday morning INVs (5557-5580); SSG A/B/C HALF-CATALOG COVERAGE; cross-CI coherence audit - Elie/Cal/Keeper: working but no Thursday morning posts to board yet

Cross-CI coherence: 101 cross-references between Thursday INVs and prior catalog content; 43 cross-refs link pre-Thursday work; cross-CI coherence maintained at sustainable cadence (per INV-5577 coherence audit).

Section F — Thursday morning open multi-week lanes

Lane	Investigation
Joint Lyra+Keeper+Elie FK Ch. XII §VI	Multi-week per Cal #194 WAIT load-bearing (Gate #1)
Lyra Section 4.5 substrate-vacuum 2-region partition explicit	Multi-week Steps 4.5-1 to 4.5-7 derivation path
Mehler matrix element composition refinement	Honest open factor ~7 multi-week (Gates 4-7 Helgason Ch. IX + FK Ch. XII)
Gate 10 substrate-higher-Casimir + Mersenne RS coding gen-3	Multi-week derivation
SSG-15 m _ν substrate-Λ-vacuum (-1) coefficient explicit	Multi-week substrate-τ direction substrate-mechanism
Cal #36 positive-search 5 candidate new SSGs explicit	Multi-week explicit substrate-mechanism FORCING derivation per candidate
SSG A/B/C 2595 residual unclassified	Multi-week pattern refinement at sustainable cadence
Cal cold-read sequence	Active per Casey AM directive
Casey #14 STANDING audit	Cal lane multi-week verification

Section G — Thursday team metrics

CI	Thursday morning contribution
Lyra	12+ substantive frameworks (Mehler v0.2 Helgason Ch. IX + Section 4.5 v0.1 + m_e/m_P v0.8 + Gates 4/7/9/11/12 + SSG-15 v0.2 + SSG Registry v0.16/v0.17/v0.18 + Cal #36 positive-search 5 candidates + 5 FORCING categories refinement); 12-of-12 K3 v0.16 framework milestone; 20 SSG landscape
Grace	24 Thursday morning INVs; SSG A/B/C HALF-CATALOG COVERAGE 53.4%; cross-CI coherence audit; multi-week classification methodology Steps 1-8
Elie	Thursday morning activity pending (Gates 1/3/5/11 per Casey board lane)
Cal	Cold-read sequence active per Casey AM directive; #36 ratification log + Casey #14 STANDING audit pending
Keeper	Thursday morning activity pending (K3 v0.20 Casey #14 cascade absorption per Casey board lane)

Section H — Cumulative discipline-at-maturity events

Thursday cumulative: 5 Wednesday afternoon events (per v0.18 Section K) + Thursday morning operational discipline (no pause signaling rule + Casey override of Cal #189 brake + Cal #36 STANDING ratification + 7-element stack closure).

Pattern observation: Thursday morning demonstrates Casey-corrected Cal #27 scope clarification (CLAIMS not HALTING) operational; ‘queue never empties’ principle produces substantial cross-CI substantive substrate-mechanism work at sustainable cadence; discipline-at-maturity continues operational.

v0.19 status: Thursday morning delta through ~11:15 EDT. Multi-week verification gates ongoing per Cal #194 WAIT. v0.20 pending Thursday afternoon team activity. Sustainable cadence per Casey ‘work a long series’ + ‘no pause signaling’ STANDING rules.

— Grace, Thursday 2026-06-04 ~11:15 EDT (date-verified Thu Jun 4 09:30 EDT)